

Section 15.2

1. Evaluate $\iint_D (x + y) dA$, where D is the triangular region with vertices $(0, 0)$, $(2, 0)$, and $(0, 4)$.

Hint: Sketch the region. The hypotenuse has equation $y = 4 - 2x$. Set this up as a Type 1 integral.

Solution:

The region is Type 1: for fixed $x \in [0, 2]$, y goes from 0 to $4 - 2x$.

$$\int_0^2 \int_0^{4-2x} (x + y) dy dx$$

Inner integral:

$$\begin{aligned} \int_0^{4-2x} (x + y) dy &= \left[xy + \frac{y^2}{2} \right]_0^{4-2x} = x(4 - 2x) + \frac{(4 - 2x)^2}{2} \\ &= 4x - 2x^2 + \frac{16 - 16x + 4x^2}{2} = 4x - 2x^2 + 8 - 8x + 2x^2 = 8 - 4x \end{aligned}$$

Outer integral:

$$\int_0^2 (8 - 4x) dx = [8x - 2x^2]_0^2 = 16 - 8 = \boxed{8}$$

2. Evaluate $\iint_D x dA$, where D is the region bounded by $x = y^2$ and $x = 4$.

Hint: Sketch the region. For a fixed y , where does x start and stop? What are the y -bounds?

Solution:

The curves intersect where $y^2 = 4$, so $y = \pm 2$.

This is naturally Type 2: for fixed $y \in [-2, 2]$, x goes from y^2 to 4.

$$\int_{-2}^2 \int_{y^2}^4 x dx dy$$

Inner integral:

$$\int_{y^2}^4 x dx = \frac{x^2}{2} \Big|_{y^2}^4 = \frac{16}{2} - \frac{y^4}{2} = 8 - \frac{y^4}{2}$$

Outer integral: By symmetry (even function of y), we can write $2 \int_0^2$:

$$2 \int_0^2 \left(8 - \frac{y^4}{2} \right) dy = 2 \left[8y - \frac{y^5}{10} \right]_0^2 = 2 \left(16 - \frac{32}{10} \right) = 2 \left(\frac{128}{10} \right) = \boxed{\frac{128}{5}}$$

3. Evaluate $\iint_D 2x \, dA$, where D is the region bounded by $x = y^2 - 1$ and $x = 3 - y^2$.

Hint: Sketch the region. Can you describe it as Type 1? As Type 2? Only one of these works as a single integral.

Solution:

Intersections: $y^2 - 1 = 3 - y^2$ gives $2y^2 = 4$, so $y = \pm\sqrt{2}$.

Why Type 2? The left boundary is $x = y^2 - 1$ and the right boundary is $x = 3 - y^2$. Both are functions of y , so for a fixed y , x goes from $y^2 - 1$ to $3 - y^2$. This is naturally Type 2.

As Type 1, the region would require splitting into multiple integrals because the left and right boundaries change at different x -values. Type 2 handles it in one integral.

$$\int_{-\sqrt{2}}^{\sqrt{2}} \int_{y^2-1}^{3-y^2} 2x \, dx \, dy$$

Inner integral:

$$\int_{y^2-1}^{3-y^2} 2x \, dx = x^2 \Big|_{y^2-1}^{3-y^2} = (3-y^2)^2 - (y^2-1)^2$$

Expand: $(9 - 6y^2 + y^4) - (y^4 - 2y^2 + 1) = 8 - 4y^2$

Outer integral: By symmetry (even function of y):

$$\begin{aligned} 2 \int_0^{\sqrt{2}} (8 - 4y^2) \, dy &= 2 \left[8y - \frac{4y^3}{3} \right]_0^{\sqrt{2}} = 2 \left(8\sqrt{2} - \frac{4(2\sqrt{2})}{3} \right) = 2 \left(8\sqrt{2} - \frac{8\sqrt{2}}{3} \right) \\ &= 2 \cdot \frac{16\sqrt{2}}{3} = \boxed{\frac{32\sqrt{2}}{3}} \end{aligned}$$

4. The integral $\int_0^4 \int_{\sqrt{y}}^2 \cos(x^3) \, dx \, dy$ cannot be evaluated directly because $\cos(x^3)$ has no elementary antiderivative in x .

Reverse the order of integration and evaluate.

Hint: Sketch the region of integration first. The inner limit $x = \sqrt{y}$ is the same as $y = x^2$.

Solution:

Original region: $0 \leq y \leq 4$ and $\sqrt{y} \leq x \leq 2$.

Rewrite $x = \sqrt{y}$ as $y = x^2$. The region is bounded on the left by $y = x^2$, on the right by $x = 2$, below by $y = 0$, above by $y = 4$.

Reverse (Type 1): For fixed $x \in [0, 2]$, y goes from 0 to x^2 :

$$\int_0^2 \int_0^{x^2} \cos(x^3) \, dy \, dx$$

Inner integral (in y , with x fixed — $\cos(x^3)$ is a constant!):

$$\int_0^{x^2} \cos(x^3) \, dy = x^2 \cos(x^3)$$

Outer integral: Let $u = x^3$, $du = 3x^2 \, dx$:

$$\int_0^2 x^2 \cos(x^3) \, dx = \frac{1}{3} \int_0^8 \cos(u) \, du = \frac{1}{3} [\sin u]_0^8 = \boxed{\frac{\sin 8}{3}}$$

(Numerically ≈ 0.3297 .)